Capstone release review

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask which gate blocks and whether its evidence matches the candidate.

Teach mandatory conditions and scoped review eligibility without invented tests, sign-offs or release actions.

Keep one habit: Model threshold equality → inspect missing rehearsal → contrast deletion and access failures → check versioned repair evidence → preserve decision history.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of capstone **release review**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Capstone failure testing. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of capstone failure testing

Reference example, not the only acceptable wording: Teacher explanation: Capstone failure testing checks denied roles, missing **evidence**, invalid **input**, timeouts and stale data. Each expected outcome is written before the run.

Use a failure matrix linking condition to expected response and forbidden calls. Test both visible result and operation trace. Preserve failed fixtures for regression checks.

Without opening the old lesson, explain one key idea from Documentation and operational readiness. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of documentation and operational readiness

Reference example, not the only acceptable wording: Teacher explanation: Operational documentation tells another person how to run, verify, observe and recover the service. A successful developer demo is not readiness.

A runbook names versions, required non-secret configuration, secret key names, health checks, release/rollback dependencies, alerts and owners. Test it with a fresh operator and fictional fixtures.

Without opening the old lesson, explain one key idea from Performance and cost experiments. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of performance and cost experiments

Reference example, not the only acceptable wording: Teacher explanation: Performance experiments compare matched workloads with declared timing boundaries and quality constraints. Averages can hide slow cases.

Use the same cases and environment; record every duration and success. For a simple median sort values and average the middle pair for even counts. Keep failed requests visible.

Pictured end goal

Explain capstone **release review** using the deciding step in this fictional service.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Capstone failure testing

Documentation and operational readiness

Performance and cost experiments

Use the three-session teaching cycle above, with extra time as needed. **Release review** combines correctness, access, failure recovery and operational readiness. **Critical gate**s can block release independently of an average score. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of capstone **release review**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach all-mandatory review with explicit evidence states and versions

> Separate main untested recovery from the picture's deletion gate and selected access failure.

Definitions and complete fictional gate contract

Release review combines correctness, access, failure recovery and operational readiness. **Critical gate**s can block release independently of an average score.

Toy release requires all access tests, no unauthorised effects, at least 90% ordinary-case pass and a compatible **rollback rehearsal**. Record each gate separately with versioned **evidence**.

Review contract	Complete fictional rule
Main candidate	r2; evidence records must match the reviewed candidate
Ordinary cases	At least 90%; main 18/20 exactly meets the threshold
Access	All declared access tests must pass
Effects	No unauthorized effects in the declared trace checks
Recovery	Compatible rollback rehearsal must have passing evidence; not run blocks
Evidence record	Gate state pass/fail/not_run, candidate version and reference for any supplied pass/fail
Separate picture	p2 requires factual, access and deletion gates; factual/access pass, listed backup residual makes deletion fail
Repair review	New p3 evidence must cover the current candidate; stale p2 passes cannot silently count for p3
Decision scope	Eligibility under supplied fictional gates, not actual deployment or human approval
Limits	Constructed evidence IDs/states are not executed checks; listed-copy verification is not unknown-copy deletion proof

Main gate ledger and checked reasoning

Main r2 gate	Supplied evidence	Gate judgment
Ordinary cases	18/20 = 90%	Pass at inclusive 90% threshold
All access tests	Supplied passing fixture record	Pass within declared cases
No unauthorized effects	Supplied clean fixture trace	Pass within declared trace
Compatible rollback rehearsal	Not run	Missing required evidence; blocks eligibility
Overall main review	Three supplied passes cannot fill rehearsal gap	Not ready under all-mandatory rule

Main 18/20 = 90% meets the ordinary threshold, access/no-effects supplied fixtures pass, but compatible rollback not run blocks. Picture p2 factual/access passes cannot outweigh deletion failure from listed backup residual. Added p3 stale-evidence example blocks until candidate applicability/current evidence is supplied; all new matching hypothetical passes mean only eligibility under the declared picture gates.

Evidence IDs are classroom inputs, not performed checks or sign-offs. Preserve the first failure and each review. Selected 27/30 = 90% ordinary passes, but one of five access tests fails, so access gate blocks despite rollback pass/no effects. Keep exact failed access case before repair.

Capstone release review — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Ordinary cases 18/20 pass; access all pass; no effects; rollback not tested.	Toy release requires all access tests, no unauthorised effects, at least 90% ordinary-case pass and a compatible rollback rehearsal.	90% ordinary threshold met but rollback evidence missing, so release is not ready. Absence of a failure report is not a passed rehearsal.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: 90% ordinary threshold met but rollback **evidence** missing, so release is not ready. Absence of a failure report is not a passed rehearsal.

Qualification: Release readiness is scoped to the declared gates and environment; deployment still needs operational checks.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Ordinary cases 18/20 pass; access all pass; no effects; rollback not tested.

90% ordinary threshold met but rollback **evidence** missing, so release is not ready. Absence of a failure report is not a passed rehearsal.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s. The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain only authorized, minimized versioned review **evidence** without secret values or unnecessary private content so a later revision can be compared honestly.

Missing/wrong-version evidence and assurance probes

Release-review probe	Expected interpretation
Ordinary rate exactly 90%	Meets the at-least threshold
All other gates pass but rollback not run	Not eligible; missing rehearsal evidence
Two passes and mandatory deletion failure	Blocked; no majority vote
Pass/fail state has no evidence reference	Reject incomplete supplied evidence record
Old candidate’s pass reused after candidate changes	Wrong-version evidence blocks current review until applicability is established
All listed locations newly pass	Scoped verification; unknown stores still outside claim
All declared current gates pass	Eligible under that fictional gate set, not automatic deployment or human sign-off
Repair succeeds on a new review	Preserve original failed case and earlier decision
Approval granted in a real process	Still no guarantee of zero future faults; retain ownership, monitoring and recovery

Separate listed-backup failure and repaired-candidate scope

Review every mandatory gate. A capstone release needs factual, access and deletion gates to pass.

Factual and access checks pass. Deletion fails because a listed backup remains. All three gates are mandatory.

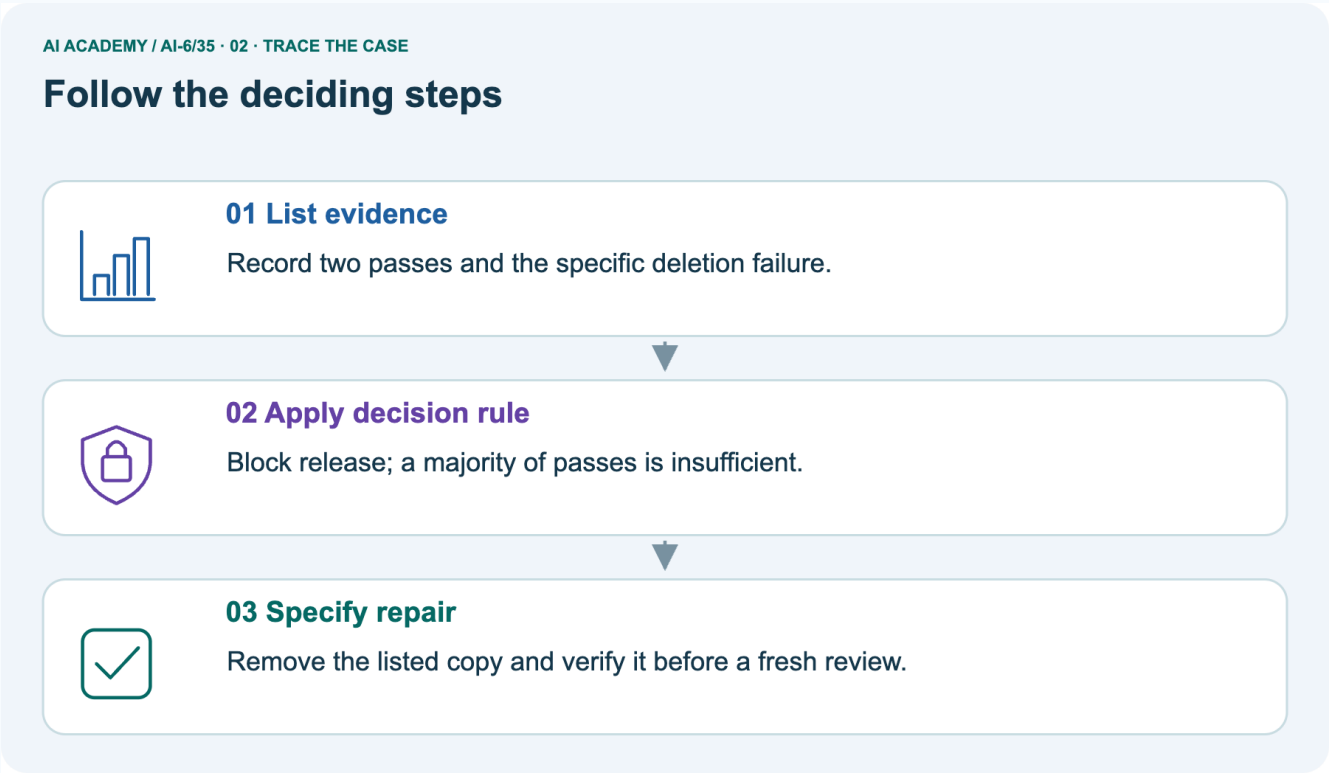
The picture has a separate three-gate contract: factual, access and deletion. Candidate p2 passes factual/access but fails deletion because a listed backup copy remains. Two passes cannot outweigh that mandatory failure.

A repaired p3 candidate needs new verification for every listed deletion location and applicable current factual/access evidence; the local code illustrates stale p2 records blocking p3, then separately supplied matching p3 passes. All states are hypothetical fixture

inputs, not real checks or deleted backups. Preserve the original failure and each later review.

If all three current pictured gates pass, that satisfies only the declared fictional gate set; it does not automatically deploy, grant human approval or guarantee zero future faults.

Picture trace and fresh-verification key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. List evidence	Record two passes and the specific deletion failure.
2. Apply decision rule	Block release; a majority of passes is insufficient.
3. Specify repair	Remove the listed copy and verify it before a fresh review.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Release review applies explicit conditions to a specific candidate version. Completed coding is not **evidence** that all operational obligations are satisfied.

Approval cannot guarantee zero future faults; retain ownership, monitoring and recovery arrangements.

Change: Every listed deletion location passes new verification.

Prediction: That gate can pass; review the complete current **evidence** before approving release.

Reason: One repaired gate does not automatically refresh **evidence** for every other gate.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Release review	A decision using versioned gate evidence
Critical gate	A requirement that cannot be averaged away
Rollback rehearsal	A tested recovery of a compatible version set

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Capstone release review

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Ordinary cases 18/20 pass; access all pass; no effects; rollback not tested. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: 90% ordinary threshold met but rollback **evidence** missing, so release is not ready. Absence of a failure report is not a passed rehearsal.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Capstone **release review**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Release readiness is scoped to the declared gates and environment; deployment still needs operational checks.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Apply the three mandatory release gates to the recorded results. Why do two passes not outweigh the deletion failure, and what **evidence** must follow its repair?

Record two passes and the specific deletion failure. Block release; a majority of passes is insufficient. Remove the listed copy and verify it before a fresh review.

Changed case: That gate can pass; review the complete current **evidence** before approving release. One repaired gate does not automatically refresh **evidence** for every other gate.

Changed-case reasoning: That gate can pass; review the complete current **evidence** before approving release. One repaired gate does not automatically refresh **evidence** for every other gate.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main ordinary threshold met but compatible rollback not run blocks. Picture listed-backup deletion failure blocks despite two passes; repaired-candidate review requires applicable current evidence and preserved history. Selected 27/30 = 90% ordinary passes, but one of five access tests fails, so mandatory access gate blocks regardless of rollback pass/no effects.

Preserve exact failed case, repair trusted enforcement and obtain fresh review evidence; no automatic approval or deployment is implied.

Diagnose the current misconceptions

The worked result establishes every future case: Release readiness is scoped to the declared gates and environment; deployment still needs operational checks.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Ordinary 27/30 pass; access 1/5 fails; rollback passes; no effects. Decide.

Teacher explanation: 90% ordinary passes, access gate fails. Block release and preserve exact failed case before repair.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: **Release review** combines correctness, access, failure recovery and operational readiness. **Critical gate**s can block release independently of an average score.

Toy release requires all access tests, no unauthorised effects, at least 90% ordinary-case pass and a compatible **rollback rehearsal**. Record each gate separately with versioned **evidence**.

W2 Trace the worked case

Ordinary cases 18/20 pass; access all pass; no effects; rollback not tested. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: 90% ordinary threshold met but rollback **evidence** missing, so release is not ready. Absence of a failure report is not a passed rehearsal.

W3 Audit the limit

A report omits this qualification: Release readiness is scoped to the declared gates and environment; deployment still needs operational checks.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Release readiness is scoped to the declared gates and environment; deployment still needs operational checks.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Release readiness is scoped to the declared gates and environment; deployment still needs operational checks.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: 90% ordinary threshold met but rollback **evidence** missing, so release is not ready. Absence of a failure report is not a passed rehearsal.

Qualification: Release readiness is scoped to the declared gates and environment; deployment still needs operational checks.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Ordinary 27/30 pass; access 1/5 fails; rollback passes; no effects. Decide.

Teacher explanation: 90% ordinary passes, access gate fails. Block release and preserve exact failed case before repair.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: 90% ordinary passes, access gate fails. Block release and preserve exact failed case before repair. Require intermediate reasoning and the stated scope.

Fresh retry: Ordinary 19/20, all access pass, one unauthorised send, rollback passes. Expected reasoning: 95% ordinary; side-effect gate fails. Aggregate score cannot override it.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

For Grounded Learning Service, use this week’s independent task as a rehearsal for the final **evidence** pack: Ordinary 27/30 pass; access 1/5 fails; rollback passes; no effects. Decide.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Engineering demonstration and defense. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of capstone **release review**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Ordinary cases 18/20 pass; access all pass; no effects; rollback not tested. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: 90% ordinary threshold met but rollback **evidence** missing, so release is not ready. Absence of a failure report is not a passed rehearsal.

Check every learner before discussion: Use the worked case for Capstone **release review**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Release readiness is scoped to the declared gates and environment; deployment still needs operational checks.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Every listed deletion location passes new verification. Require a **prediction** and a reason before comparing. Expected result: That gate can pass; review the complete current **evidence** before approving release.

Reason: One repaired gate does not automatically refresh **evidence** for every other gate.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Capstone release review: the case



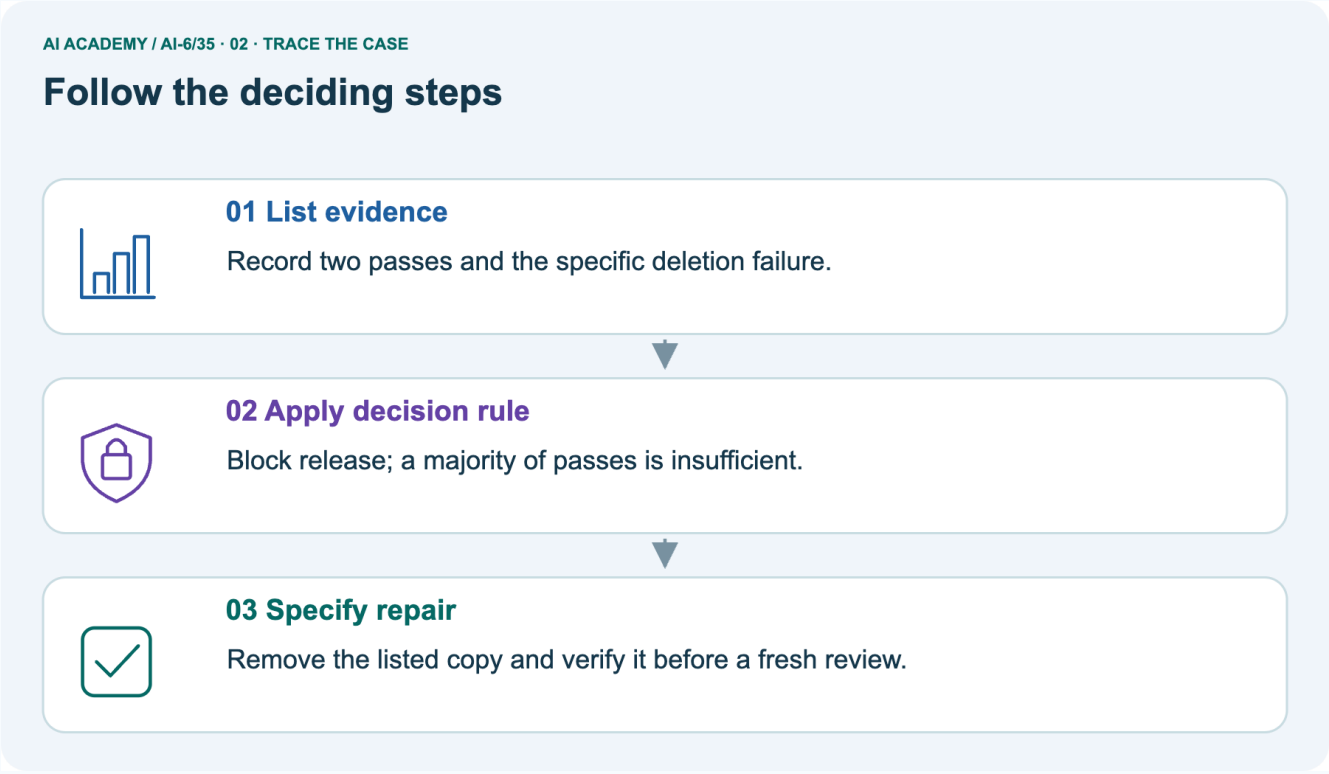
Follow the labelled picture with the matching explanation. Open the vector diagram

Review every mandatory gate. A capstone release needs factual, access and deletion gates to pass. Supplied facts: Factual and access checks pass. Deletion fails because a listed backup remains. All three gates are mandatory.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



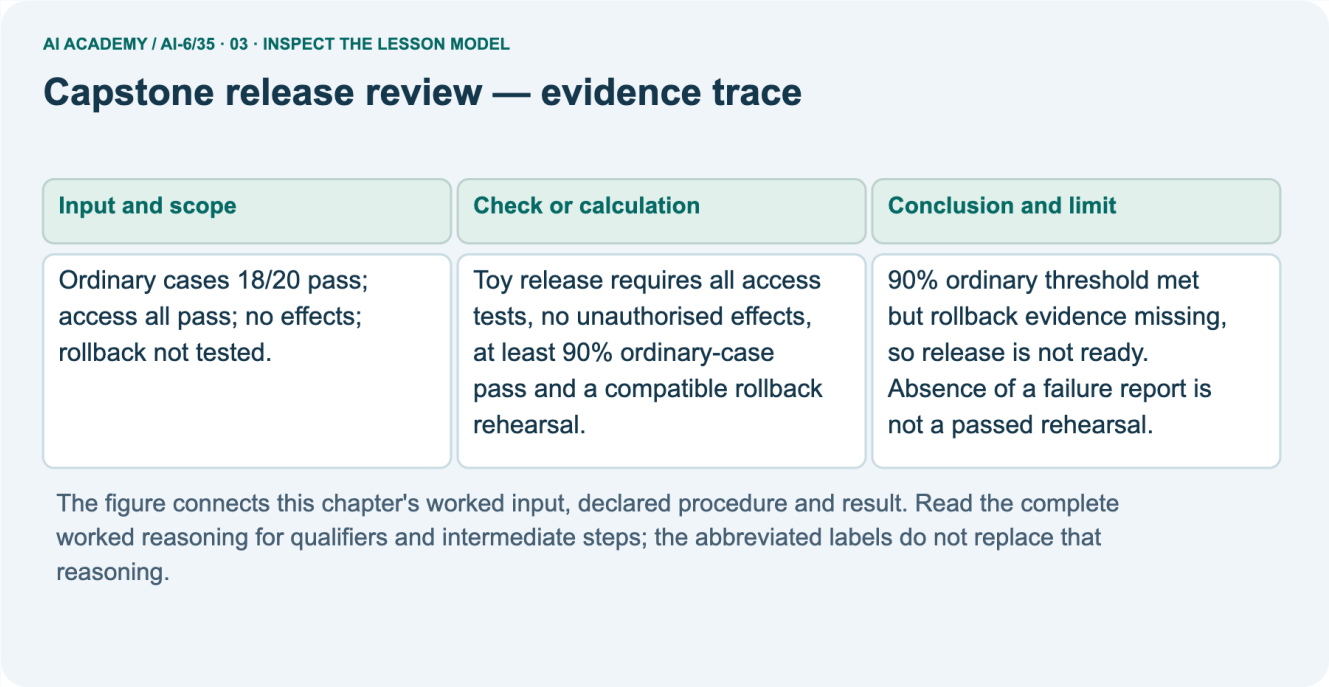
Follow the labelled picture with the matching explanation. Open the vector diagram

List **evidence**: Record two passes and the specific deletion failure. → Apply decision rule: Block release; a majority of passes is insufficient. → Specify repair: Remove the listed copy and verify it before a fresh review.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Capstone release review — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone **release review** — **evidence** trace.

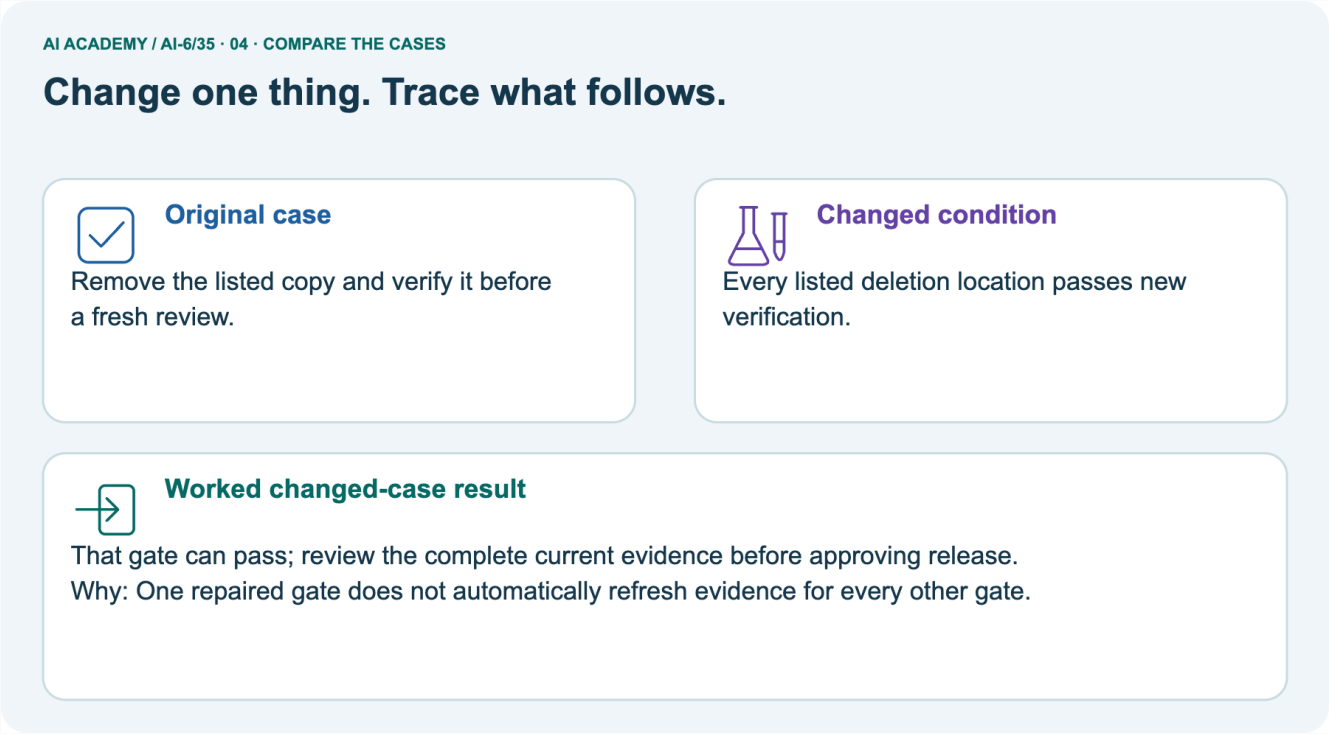
Ordinary cases 18/20 pass; access all pass; no effects; rollback not tested. | Toy release requires all access tests, no unauthorised effects, at least 90% ordinary-case pass and a compatible **rollback rehearsal**. | 90% ordinary threshold met but rollback **evidence** missing, so release is not ready.

Absence of a failure report is not a passed rehearsal.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Remove the listed copy and verify it before a fresh review. Changed condition: Every listed deletion location passes new verification.

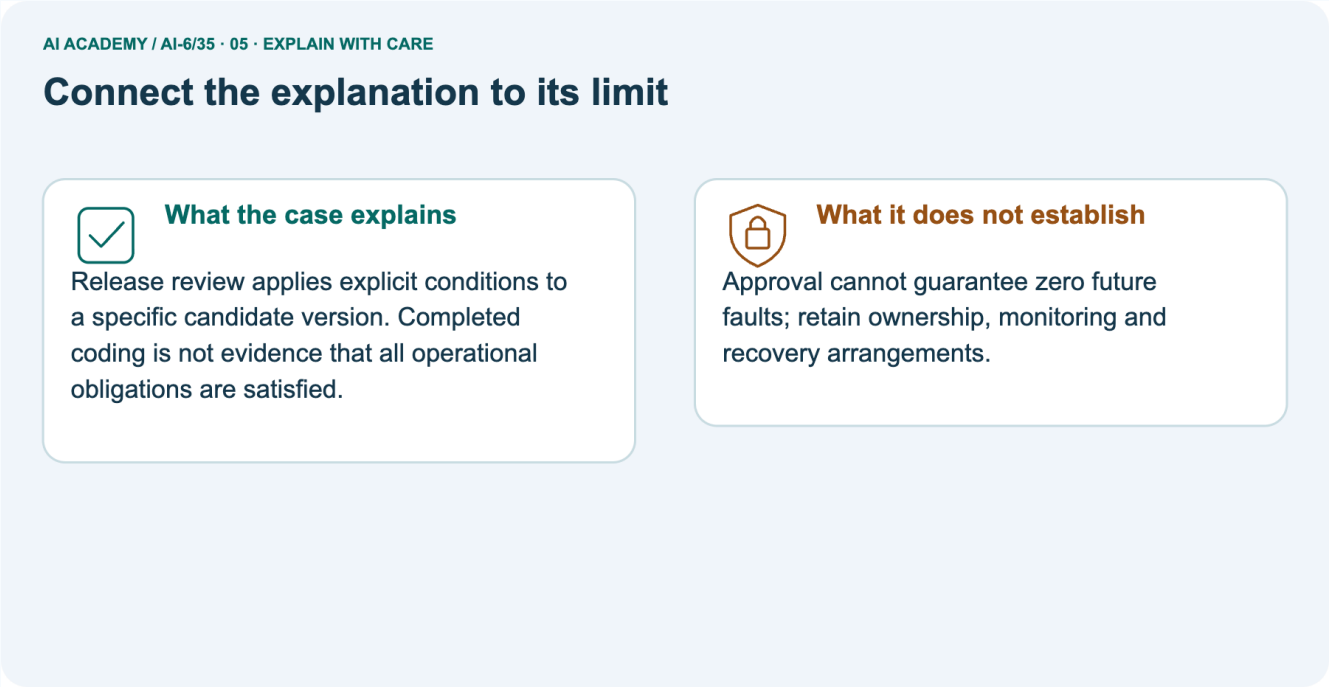
Expected result: That gate can pass; review the complete current **evidence** before approving release. Reason: One repaired gate does not automatically refresh **evidence** for every other gate.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: One repaired gate does not automatically refresh **evidence** for every other gate.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Release review applies explicit conditions to a specific candidate version. Completed coding is not **evidence** that all operational obligations are satisfied.

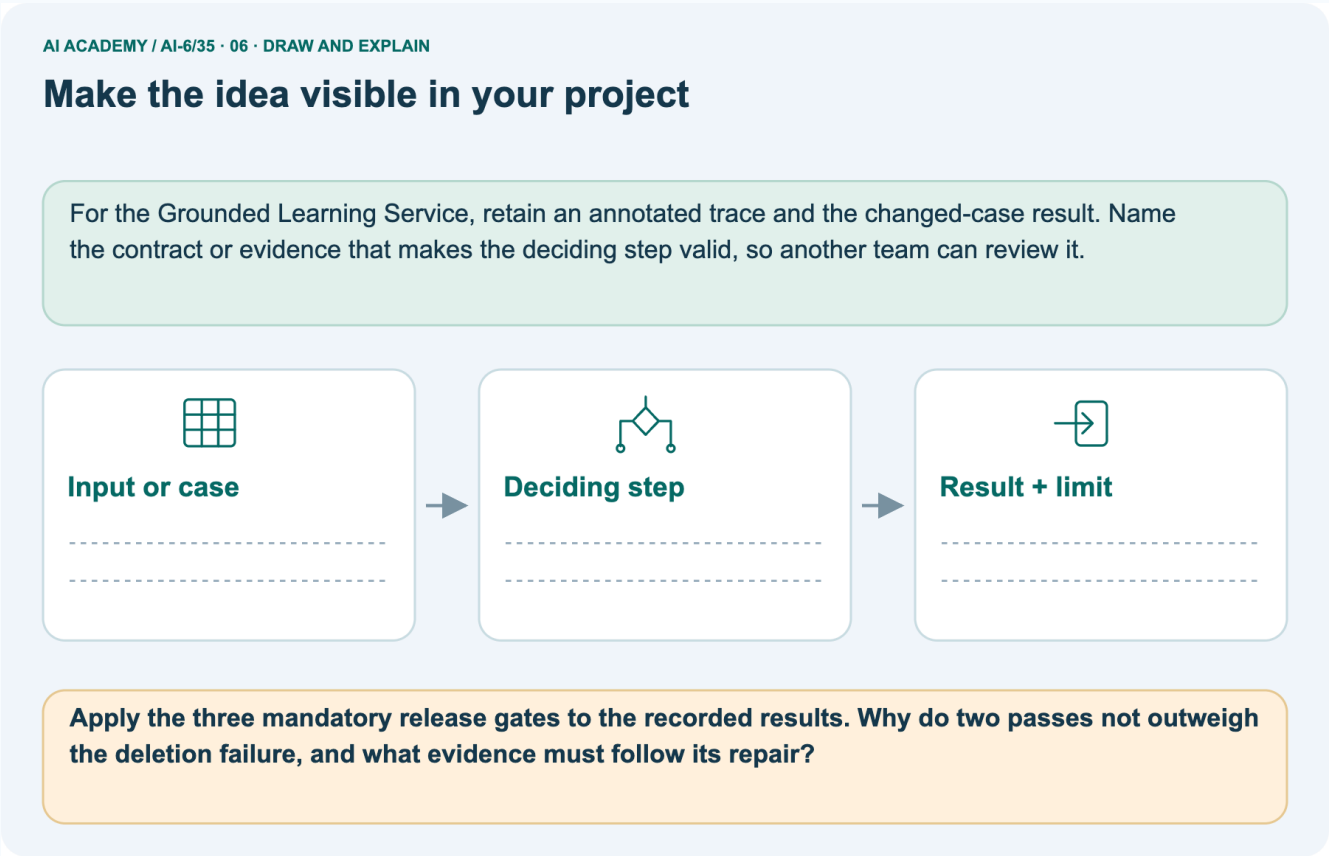
Limit: Approval cannot guarantee zero future faults; retain ownership, monitoring and recovery arrangements.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Approval cannot guarantee zero future faults; retain ownership, monitoring and recovery arrangements.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Apply the three mandatory release gates to the recorded results.

Why do two passes not outweigh the deletion failure, and what **evidence** must follow its repair?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Record two passes and the specific deletion failure. Block release; a majority of passes is insufficient. Remove the listed copy and verify it before a fresh review.

Changed case: That gate can pass; review the complete current **evidence** before approving release. One repaired gate does not automatically refresh **evidence** for every other gate.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Capstone release review: the case

02. Follow the deciding steps

03. Capstone release review — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.